

BHAVIN BALDOTA

AI / ML Engineer | GenAI & LLM Specialist | Agentic AI

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PROFESSIONAL SUMMARY

GenAI / ML Engineer with 2+ years of production experience architecting enterprise LLM systems, RAG pipelines, and Agentic AI applications. Led GenAI platform engineering at Persistent Systems — delivering multi-agent orchestration frameworks, scalable AI microservices, and LLMOps pipelines across enterprise clients. Research background with DRDO, 2 Elsevier datasets published, and papers in progress. Full GenAI stack: LangChain, LangGraph, MCP, FastAPI, PyTorch, ChromaDB, FAISS.

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, JavaScript

GenAI & LLM: LangChain, LangGraph, MCP, OpenAI API, Groq, Ollama, Fine-tuning LLMs, Prompt Engineering, Agentic AI

RAG & Vector DBs: RAG Systems, ChromaDB, FAISS, Hybrid Search, Semantic Embeddings, Chunking & Retrieval Strategies

ML / DL: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, Deep Reinforcement Learning

Backend & Cloud: FastAPI, Flask, Firebase, REST APIs, AWS, Docker, Git/GitHub, CI/CD

LLMOps: Evaluation Pipelines, Hallucination Mitigation, Guardrails, Cost Optimization, Multi-Agent Orchestration

PROFESSIONAL EXPERIENCE

Persistent Systems

Sep 2025 – Present

Lead Software Engineer – GenAI

Pune, India

- Architected 5+ enterprise GenAI systems — RAG pipelines and AI assistants handling **10,000+ daily queries** at sub-200ms latency.
- Engineered production AI microservices (FastAPI + ChromaDB/FAISS) for semantic search across 50,000+ documents at **99.5% uptime**.
- Reduced LLM hallucination by **35%** via prompt engineering and custom evaluation pipelines; cut inference cost by 25% through model optimization.
- Built LangGraph multi-agent orchestration with tool-calling and reflection loops, cutting manual workflow time by **60%**.
- Mentored 3 junior engineers**; led cross-functional delivery of 4 production AI solutions with product, data, and platform teams.

Provilac

Nov 2024 – Aug 2025

AI / ML Engineer

Pune, India

- Deployed AI solutions using Deep Learning, Computer Vision, and GenAI, improving process automation efficiency by **30%**.
- Built scalable RAG pipelines and LLM knowledge assistants achieving **88%+** accuracy on domain-specific enterprise document corpora.
- Shipped GenAI features end-to-end — data ingestion to inference — reducing research-to-production cycle by **40%**.

DRDO – Defence Research and Development Organisation

Jul 2024 – Dec 2024

Research Intern (AI / ML)

India

- Developed deep learning models for AI/ML-based audio perception and sound source localization achieving **92% localization accuracy** across multi-channel acoustic environments.

KEY PROJECTS

DeliverIQ – Delivery Intelligence System (9-Module AI/ML Suite)

IEEE Access – Submission Pending

Python · PyTorch · FastAPI · LangChain · ChromaDB · Computer Vision · RAG

- 9 interconnected AI/ML modules spanning route optimization, demand forecasting, anomaly detection, and CV-based package tracking for last-mile intelligence.
- End-to-end RAG pipeline achieving **90%+** retrieval accuracy for delivery Q&A; paper in preparation for IEEE Access with full benchmark evaluations.

Enterprise RAG-Based Document Intelligence System

LangChain · ChromaDB · FAISS · FastAPI · OpenAI API · Python

- Production RAG pipeline with recursive chunking, hybrid search, and metadata filtering; reduced document retrieval time by **70%** with 90%+ LLM answer accuracy.

Agentic AI – Multi-Agent Orchestration Framework

LangGraph · LangChain · MCP · FastAPI · ChromaDB · Python

- Multi-agent system with tool-calling, persistent memory, and reflection loops deployed for **500+ daily sessions**; cut workflow completion time by 60% vs single-agent baselines.

EDUCATION

Vishwakarma University, Pune, India

2021 – 2025

B.Tech in Artificial Intelligence and Data Science | CGPA: 9.01 / 10

Coursework: Deep Learning · NLP · Computer Vision · Reinforcement Learning · Data Structures & Algorithms · Cloud Computing

PUBLICATIONS & RESEARCH

- [Coconut Tree Disease Dataset](#) — Elsevier Data in Brief. Contributed to ML model development, training pipeline, and evaluation for plant disease classification.
- [Natural Pothole Dataset \(Abrasion & Cavitation\)](#) — Elsevier Data in Brief. Contributed to CV-based road damage detection model training and dataset curation.
- 2 research papers in progress as primary author targeting IEEE Access & Springer (2026); DeliverIQ paper in active preparation.

CERTIFICATIONS & ACHIEVEMENTS

- [TCS iON | Artificial Intelligence Pro](#) — Certified
- [60+ certifications](#) across Deep Learning, GenAI, LLMOps, AWS, Computer Vision (Coursera · DeepLearning.AI · LinkedIn Learning)
- 24+ end-to-end AI/ML project repos on GitHub ([CoderFatherBB](#)) |